# Multi-Phase Search Strategy and Results

# Three-Phase Architecture

## Three-Phase Architecture I

This review employs a **three-phase search strategy** that progressively refines coverage from broad foundational literature through technology-specific studies to targeted molecular detection analyses. Each phase builds on prior phases through cross-phase citation validation and shared deduplication.

## Phase 1: Exoplanet Atmosphere Foundation I

**Objective:** Establish the foundational literature on exoplanet atmospheric studies, covering the broadest range of research from 2010 onwards.

**Queries:** 1. `"exoplanet atmosphere" OR "exoplanet atmospheric"` 2. `"transit spectroscopy" AND "atmosphere"` 3. `"atmospheric composition" AND "exoplanet"`

**Engines:** arXiv, OpenAlex, Crossref, Semantic Scholar

**Deterministic Filters:** - Minimum year: 2010 - Maximum year: 2026 - Minimum citation count: 0 (inclusive to capture recent work)

## Phase 1: Exoplanet Atmosphere Foundation I

**Results:** 46 papers discovered, forming the foundational knowledge base for the review. This phase captures the broadest scope of atmospheric research, from observational characterizations to theoretical modeling studies.

## Phase 2: James Webb Space Telescope Studies I

**Objective:** Identify papers specifically utilizing JWST observational data for exoplanet atmospheric analysis, capturing the post-launch (2021+) surge in high-precision spectroscopic observations.

**Queries:** 1. `"James Webb" AND "atmosphere" AND ("exoplanet" OR "planet")` 2. `"JWST" AND "transit" AND "spectroscopy"` 3. `"NIRSpec" OR "MIRI" OR "NIRCam" AND "exoplanet"`

**Engines:** arXiv, OpenAlex, Crossref, Semantic Scholar

## Phase 2: James Webb Space Telescope Studies I

**Deterministic Filters:** - Minimum year: 2020 (capturing pre-launch calibration papers) - Maximum year: 2026

**Dependencies:** Builds on Phase 1 foundational papers.

**Results:** 17 papers after deterministic filtering. This phase targets records associated with JWST-era observations and analysis. Any claim about instrumental capability or scientific findings requires live, source-backed review.

## Phase 3: Molecular Detection Studies I

**Objective:** Identify papers focused on detecting and analyzing specific atmospheric molecules (H O, CO , CH , H S, Na, K) in the configured domain.

**Queries:** 1. `"water vapor" AND "exoplanet" AND ("detecti on" OR "abundance")` 2. `"carbon dioxide" AND "exoplanet " AND ("CO2" OR "atmosphere")` 3. `"methane" AND "exopla net" AND ("CH4" OR "atmosphere")` 4. `"hydrogen sulfide" OR "H2S" AND "exoplanet"` 5. `"sodium" OR "potassium" AN D "exoplanet atmosphere"`

## Phase 3: Molecular Detection Studies I

**Engines:** arXiv, OpenAlex, Crossref, Semantic Scholar

**Deterministic Filters:** - Minimum year: 2015 - Maximum year: 2026

**Dependencies:** Builds on both Phase 1 and Phase 2 papers.

**Results:** 29 papers covering molecular detection across diverse exoplanet populations, from hot Jupiters to terrestrial worlds.

# Cross-Phase Analysis

## Phase Overlap I

The three phases produced complementary but overlapping corpora, with 49.5% average Jaccard similarity between phases. Papers discovered in multiple phases are tracked with full provenance, enabling analysis of papers that span multiple research paradigms.

## Citation Validation I

Cross-phase citation analysis records that 39.4% of later-phase papers cite foundational work from Phase 1. This is a descriptive connection statistic for the retained corpus; it does not establish methodological coherence across the field.

## Combined Corpus I

The deduplicated combined phase corpus contains 46 unique papers spanning 2000–2023 (23 years). It provides a multi-phase description of the retained retrieval slice, not a field-wide account.

# LLM-Based Content Filtering

# LLM-Based Content Filtering I

Three LLM-based content filters were designed for this review:

# LLM-Based Content Filtering I

1. **Study Type Classification**: Categorizes papers as observational, theoretical, review, or other. Keeps only observational and theoretical studies for analysis.
2. **JWST Data Analysis**: Identifies papers that analyze actual JWST observational data, distinguishing from theoretical JWST predictions.
3. **Molecular Detection Focus**: Filters for papers primarily focused on detecting and measuring specific atmospheric molecules.

## LLM-Based Content Filtering I

These filters are configurable through `manuscript/config.yaml` and can be enabled for specific phases or applied across the entire corpus. When enabled, they use a local Ollama LLM instance for cost-effective, privacy-preserving content classification.